

Tensor computations

Exact best-rank-one approximation ratios for general tensor formats

Difficulty: extreme **Importance:** interesting to the community

Topic: Tensor approximation and norm comparison.

For integers $d \geq 3$ and $2 \leq n_1 \leq \dots \leq n_d$, let $V = \mathbb{R}^{n_1 \times \dots \times n_d}$. For $T \in V$, define

$$\|T\|_F^2 = \sum_{i_1=1}^{n_1} \dots \sum_{i_d=1}^{n_d} T_{i_1 \dots i_d}^2,$$

$$\|T\|_\sigma = \max_{\substack{x_j \in \mathbb{R}^{n_j} \\ \|x_j\|_2=1, 1 \leq j \leq d}} \left| \sum_{i_1=1}^{n_1} \dots \sum_{i_d=1}^{n_d} T_{i_1 \dots i_d} \prod_{j=1}^d (x_j)_{i_j} \right|.$$

Determine, as a function of the dimensions, the exact value

$$\tau(n_1, \dots, n_d) = \min_{T \in V \setminus \{0\}} \frac{\|T\|_\sigma}{\|T\|_F}.$$

Equivalently, determine the largest constant τ for which $\tau\|T\|_F \leq \|T\|_\sigma$ holds for every tensor in that format. The target is the sharp constant for general formats, not merely another upper or lower bound. All formats constitute one problem.

This ratio measures the worst-case amount of tensor energy accessible to a best rank-one approximation and is relevant to convergence analyses of successive rank-one approximation methods.

References

1. Z. Li and Y.-B. Zhao, [On norm compression inequalities for partitioned block tensors](#), *Calcolo* 57 (2020), article 11, §5, equation (15), pp. 15–16. The text explicitly identifies determination of the general extremal ratio as the outstanding problem.
2. Z. Li, Y. Nakatsukasa, T. Soma, and A. Uschmajew, [On orthogonal tensors and best rank-one approximation ratio](#), *SIAM Journal on Matrix Analysis and Applications* 39 (2018), 400–425, abstract and characterization of equality in the basic dimension bound.
3. K. Kozhasov and J. Tonelli-Cueto, [Probabilistic bounds on best rank-one approximation ratio](#), arXiv v2 (2022-09-23), abstract and bounds for general and partially symmetric tensors.

Status check — 2026-09-08

Checked the 2020 published text and the later probabilistic-bound paper’s latest listed arXiv version. Searches for “best rank-one approximation ratio 2026” and “extremal ratio tensor spectral Frobenius 2025 2026” found no exact general formula. Known orthogonal-tensor cases and probabilistic dimension bounds give partial answers. They are not additional open entries. This target differs from counting critical rank-one approximations, choosing an inner product to minimize that count, and tensor-train quasi-optimality. No recent explicit reaffirmation of the full exact-value problem was located.